IPL Spatio-Temporal Weather & Pitch Analysis

Visualization & Spatial Analysis Plan for Yashmitha (Member 2 — Weather & Location Lead)

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1. Executive Objective & Collaboration Context

Our recent 15-iteration ML benchmark verified that micro-weather signals (specifically **Dew Spread**, **Relative Humidity**, and **Ambient Temperature**) are among the strongest real-world drivers of bowler performance. To align Member 2 (Weather & Location) with Member 4 (ML & Results) for our joint final presentation and research paper, this plan outlines the exact visual deliverables Yashmitha can generate from the weather and GPS datasets.

2. Core Recommended Visualizations for Member 2

Plot Type / Deliverable	Variables to Plot	Key Domain Story & Scientific Rationale
1. Geospatial Stadium Dew Map (Interactive Folium / Cartopy Map)	• Venue GPS (Lat, Lon) • Mean Dew Spread (°C) • Night Match Humidity (%)	Color-code stadium markers by mean dew intensity. Highlights high-risk coastal venues (Chennai, Mumbai) vs. dry inland venues (Jaipur, Delhi).
2. Hourly Weather Climatology Heatmap (Seaborn Matrix)	• Match Month (March-May) • Hour of Day (16:00-23:00) • Relative Humidity / Dew Point	Shows rapid moisture buildup starting at 19:30–20:00 IST (2nd innings), explaining the wet-ball grip loss found in Member 4's ML models.
3. Venue Climatology Radar Chart (Multi-Axis Comparison)	• Temperature • Relative Humidity • Wind Speed • Dew Risk Factor • Elevation (m)	Side-by-side radar profiles for **Chinnaswamy (Bengaluru)** vs. **Chepauk (Chennai)** vs. **Wankhede (Mumbai)**, demonstrating contrasting micro-climates.
4. Dew Condensation Threshold Curve (KDE / Bivariate Scatter)	• Ambient Temperature (°C) • Dew Point (°C) • Dew Spread = Temp - DewPoint	Plot the condensation boundary line (Spread ≤ 6°C) where ball grip degrades, visually supporting Member 4's EBM shape function.
5. Multi-Season Weather Trend Lines (FacetGrid Time Series)	• IPL Seasons (2008–2026) • Venue April/May Temperature • Humidity Spikes across Years	Illustrates climate variation across the 18 IPL seasons and validates sensor data consistency and absence of missing values.

3. Specific Venue Case Study for the Viva / Paper

In Section 6 of Member 4's ML report, we contrasted **M. Chinnaswamy (Bengaluru)** and **MA Chidambaram (Chepauk, Chennai)**. Yashmitha's visual slides will be the anchor for this section:

- **Chepauk (Chennai):** Mean Humidity = **72.9%**, Mean Dew Point = **24.4°C**, Mean Dew Spread = **5.87°C** (frequently inside the high condensation zone). Her plot should highlight Chennai's high marine moisture.
- **Chinnaswamy (Bengaluru):** Mean Humidity = **56.1%**, Mean Dew Point = **15.7°C**, Elevation = **~920m**, Mean Wind Speed is elevated. Her plot should illustrate the high altitude, drier evening air, and wind vectors.

4. Recommended Python Libraries & Export Specifications

- **Libraries:** matplotlib, seaborn, folium (or plotly for interactive HTML maps), and geopandas.
- **Resolution:** Export all static figures as **300 DPI PNGs** or vector **PDFs** for IEEE paper inclusion.
- **Color Palette:** Use colorblind-friendly palettes (e.g., Seaborn mako or viridis) consistent with the ML outputs.

Collaboration Note for Member 2 & 4: *Member 4's models verified that when Dew Spread drops below 6°C, pace bowler economy increases by +0.86 RPO and spinners concede +1.14 RPO due to ball slipperiness. Combining Yashmitha's spatial weather maps with Akilan's ML predictions gives our team a cohesive, publication-grade presentation!*